

Chen Qian

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Summary

- **Applied Mathematics Ph.D. candidate at UC Davis** with an **M.S. in Statistics**; research interests in nonparametric time series, stochastic processes, asymptotic theory, and statistical inference for dependent data.
- Developing **sieve-based estimation and simultaneous inference** for time-varying mean, autocovariance, and autocorrelation in locally stationary processes, with emphasis on uniform convergence theory.
- Author of a paper published in **JRSSB** that develops a theoretically rigorous classifier based on second-order characteristics and establishes asymptotically vanishing misclassification error under mild class separation.

Education

University of California, Davis <i>Ph.D. in Applied Mathematics</i>	Expected 2028 <i>Davis, CA</i>
University of California, Davis <i>M.S. in Statistics</i>	June 2023 <i>Davis, CA</i>
Complutense University of Madrid <i>B.S. in Applied Statistics</i>	July 2020 <i>Madrid, Spain</i>

Selected Research Projects

Simultaneous Inference for Locally Stationary Time Series	Sept. 2025 – Present
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- Developed sieve estimators for time-varying trend, autocovariance, and autocorrelation using orthonormal basis expansions and ordinary least squares.
- Established uniform consistency and explicit convergence rates under polynomially decaying physical dependence, including admissible regimes in which the lag increases with the sample size.
- Developing Gaussian multiplier-bootstrap simultaneous confidence bands for time-varying dependence over the full time domain, including boundary regions.

Structural Classification of Locally Stationary Time Series	Nov. 2024 – Apr. 2026
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- Developed a classification framework for locally stationary time series based on time-varying second-order structure and autoregressive approximation.
- Established conditions under which the misclassification error converges to zero, without requiring the training sample size to increase.
- Verified the finite-sample theory through simulations and real data, achieving up to **94.7% test accuracy** and outperforming statistical and deep-learning benchmarks.

Time-Frequency Methods for Nonstationary Signals	Apr. 2025 – June 2025
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- Studied Pseudo Wigner–Ville distributions and wavelet packet transforms as alternative representations of localized time-frequency structure in nonstationary signals.
- Compared their statistical resolution, predictive performance, and computational cost in financial forecasting and biomedical classification using random forests and convolutional neural networks.
- Achieved **74.7% mean directional accuracy** under the best financial-model configuration and **81.3% biomedical classification accuracy**, while identifying a substantial accuracy–runtime tradeoff between the two representations.

Publication

- **Qian, C., Ding, X., & Li, L. (2026).** Structural classification of locally stationary time series based on second-order characteristics. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, qkag083. [DOI](#)

Technical Skills

Programming: Python, R, SQL

Languages: Mandarin Chinese (native), English (professional), Spanish (working proficiency)

Teaching Experience

University of California, Davis

2023 – Present

Teaching Assistant

Davis, CA

- Led discussion sections and office hours for courses in probability, statistics, data science, calculus, and applied mathematics; prepared instructional materials and supported assessment.

University of California, Davis

July 2026

COSMOS summer program Teaching Fellow

Davis, CA

- Taught academically advanced high-school students core ideas in statistical modeling, exploratory analysis.
- Mentored student teams in formulating research questions, selecting methods, interpreting results, and presenting technically sound conclusions.

Additional Professional Experience

Zhongyu Tech

Sept. 2020 – May 2021

Data Scientist Intern, Mobile Game Analytics

Wuhan, China

- Designed product experiments and conducted cohort, retention, and churn analyses for the mobile game *Wheel Offroad 3D*.
- Built ETL workflows and recurring analytical reports that translated gameplay data into recommendations for product and engineering teams.

Honors & Awards

- 2026 L&S Graduate Summer Support Award, University of California, Davis (selected; declined due to COSMOS Teaching Fellow appointment).